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WPC POWER DRIVER BOARD – COST–OPTIMISED MODERN VERSION – functional replacement for Williams A–12697–3 (same outline, connectors, fuses).
DPAK MOSFET solenoid/lamp drivers, TPS54360B buck regulators for +5V/+12V, 4–quadrant insulated triacs, SMD passives, 5x20 mm fuses.
Sheets: Power_Supply, CPU_Interface, Sol_HighPower (1–8), Sol_LowPower (9–16), Sol_Flashers (17–24), Sol_GeneralPurpose (25–28),
Lamp_Columns, Lamp_Rows, GI_Triacs. Every part carries Manufacturer / MPN / DigiKey link / Price / Price100 fields (see BOM). See README.md.

Power_Supply

File: power_supply.kicad_sch

CPU_Interface

File: cpu_interface.kicad_sch

Sol_HighPower

File: sol_highpower.kicad_sch

Sol_LowPower

File: sol_lowpower.kicad_sch

Sol_Flashers

File: sol_flashers.kicad_sch

Sol_GeneralPurpose

File: sol_gp.kicad_sch

Lamp_Columns

File: lamp_columns.kicad_sch

Lamp_Rows

File: lamp_rows.kicad_sch

GI_Triacs

File: gi_triacs.kicad_sch

Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /
File: wpc_power_driver_cost.kicad_sch

Title: WPC Power Driver Board – cost–optimised modern re–implementation

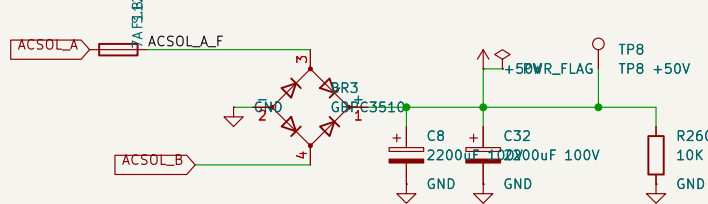
Size: A3Date: 2026–09–04Rev: A

KiCad E.D.A. 10.0.6Id: 1/10

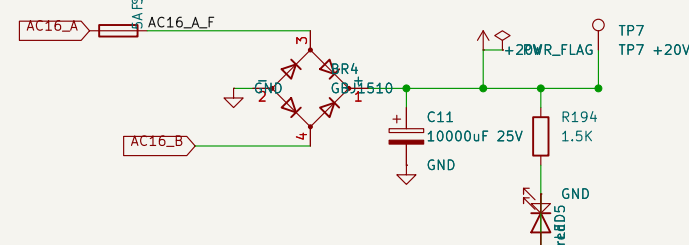
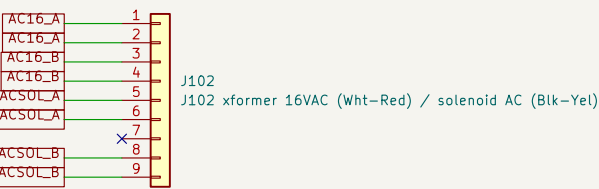
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Power supplies. Transformer secondaries enter on J101/J102/J112: each is fused (F111–F114, F116, 5x20 mm) before its bridge: GBPC3510 (BR3, +50V) and GBU1510 (BR1 +18V, BR4 +20V) on the chassis heatsink, GBU8J (BR2, +5V raw), GBU4J (BR5, 9.8VAC), +50V (BR3, 2 x 2200uF/100V, bleeder R260) feeds the solenoid outputs through F103/F104/F105. +20V (BR4/C11 10000uF) feeds flashlamps and the +12V buck U21. +18V (BR1/C6/C7 2 x 10000uF) feeds the lamp matrix columns, +5V; BR2 -> C5 (10000uF) -> TPS54360B buck U20 (3 A), +12V; +20V -> TPS54360B buck U21 (2 A) -> F115 -> J114/J116/J117. LED1 +12V, LED2 +12V fused, LED4 +5V, LED5 +20V, LED6 +18V, LED7 +12VU. TP1 12VU, TP2 5V, TP3 12V, TP5 GND, TP6 18V, TP7 20V, TP8 50V, TP9 5V raw.

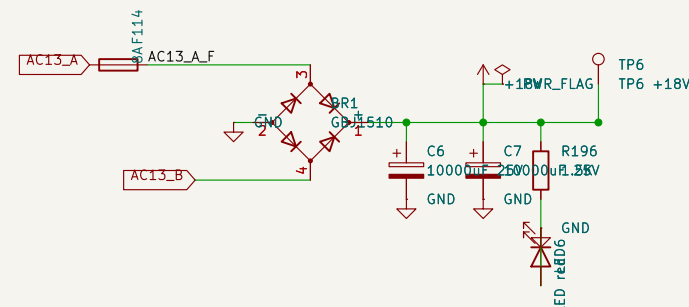
+50V solenoid supply



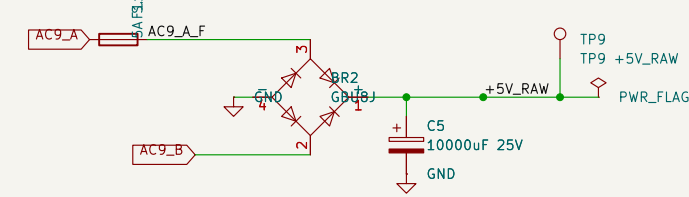
+20V flashlamp supply



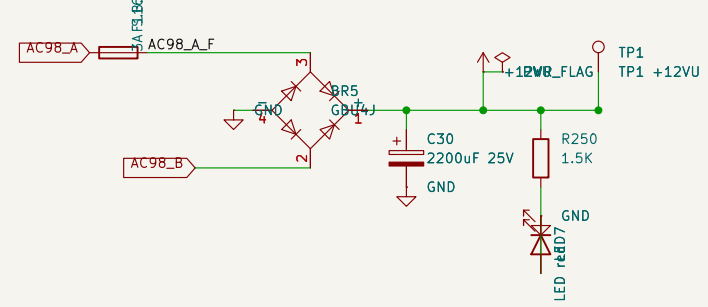
+18V lamp matrix supply



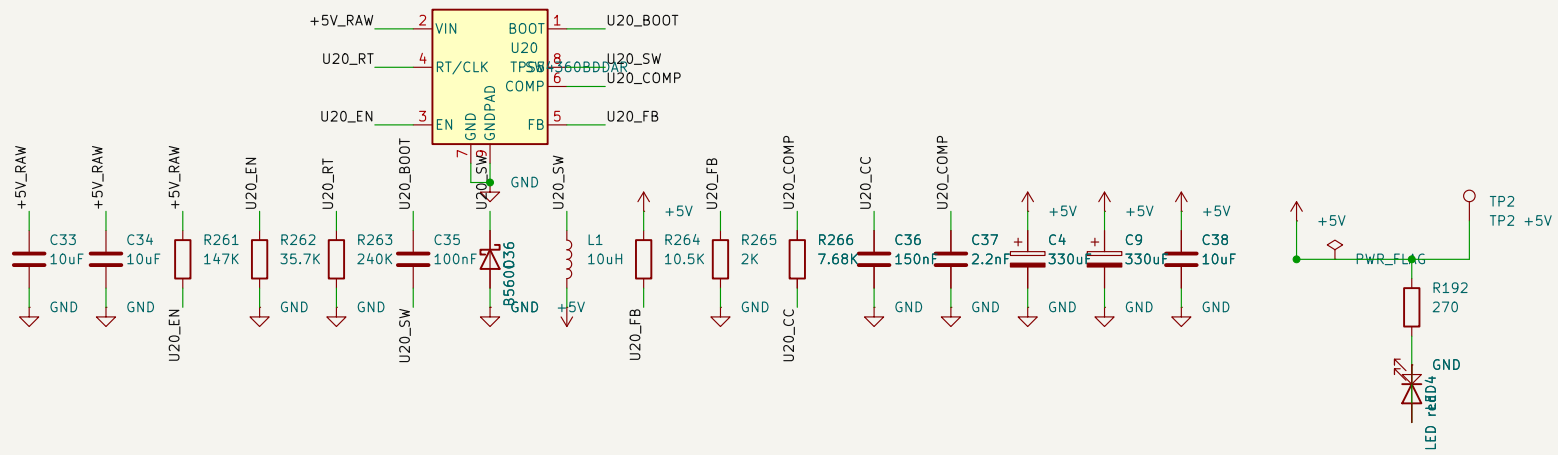
+5V raw supply (9VAC)



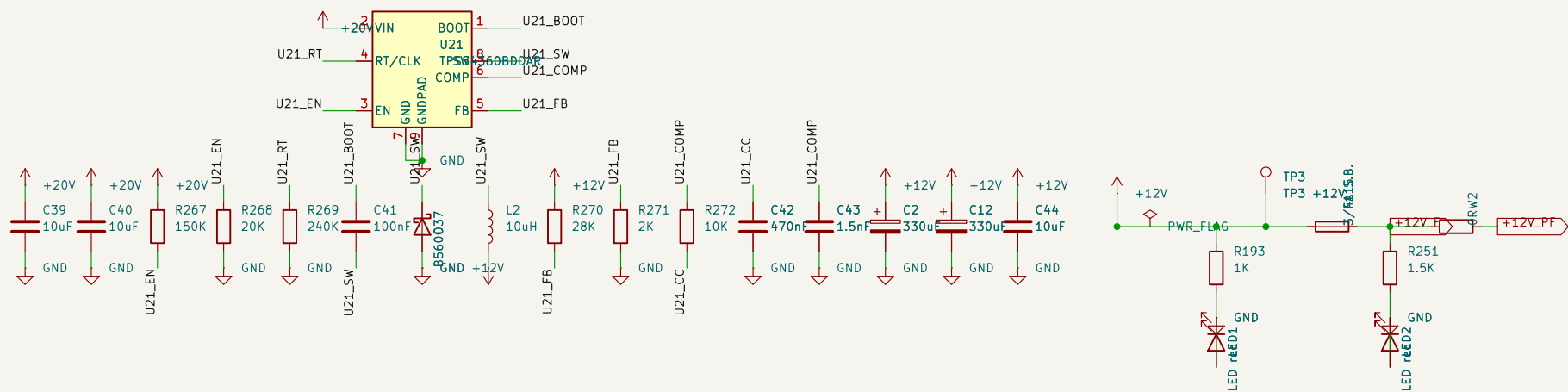
+12V unregulated / zero-cross (9.8VAC)



U20: +5V_RAW -> +5V (+5.0 V, 3 A) TPS54360B, 400 kHz, UVLO start 6.0 V / stop 5.5 V, Rc 7.68K / Cc 150nF / Cp 2.2nF



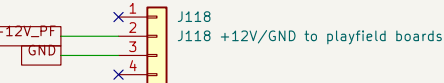
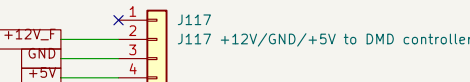
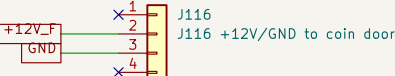
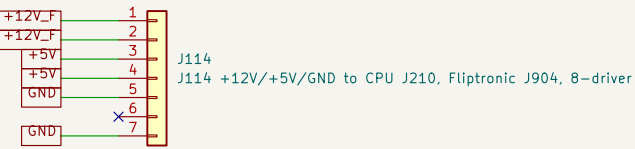
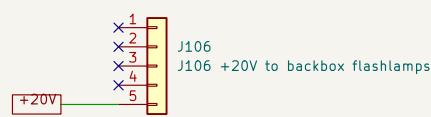
U21: +20V -> +12V (+12.0 V, 2 A) TPS54360B, 400 kHz, UVLO start 7.8 V / stop 7.3 V, Rc 10K / Cc 470nF / Cp 1.5nF



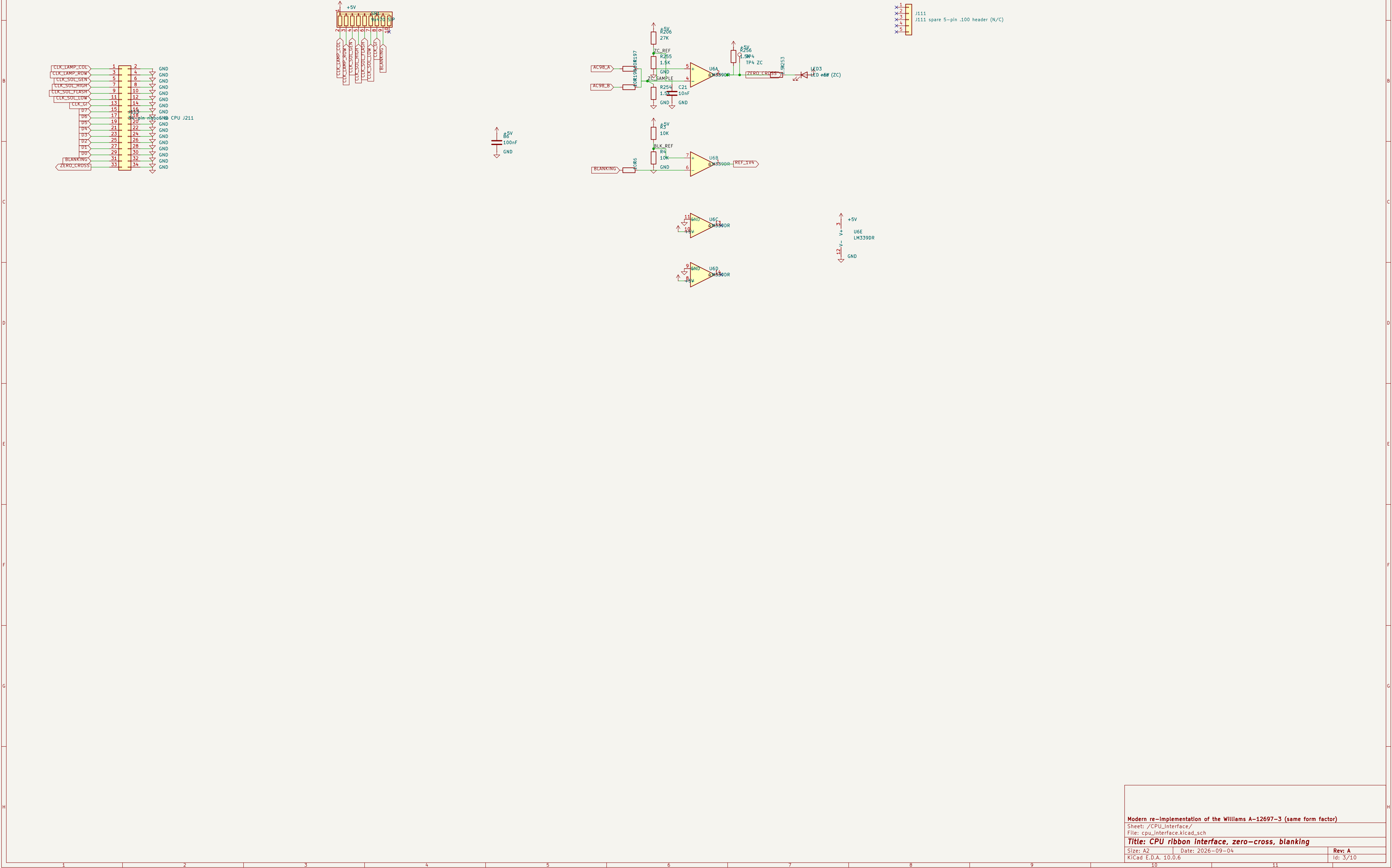
+50V

+50V

+50V

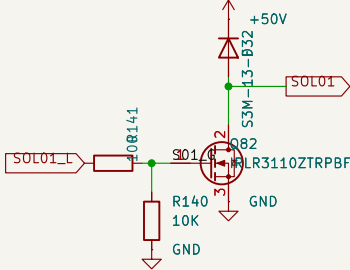


CPU interface, J113 ribbon: odd pins = signals, even pins = ground (WPC convention). D0–D7 on pins 29..15, BLANKING on pin 31, ZERO CROSS (to CPU) on pin 33, latch strobes on pins 1–13 (assignment: see README – verify against the CPU board). All latches take the TRUE data bus (bit = 1 -> output ON); BLANKING high tri-states every 74HCT574 (gate pull-downs -> all OFF) and pulls the lamp-row 1.4 V reference low. SR1 pulls strobes and BLANKING high so every driver stays OFF with the ribbon unplugged. Zero-cross: both legs of the 9.8VAC winding are summed through 10K/10K into U6A (threshold 0.26 V) -> ~1.3 ms low pulse at 120 Hz.

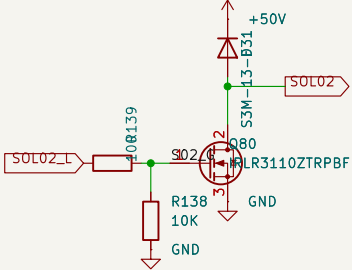


High power solenoid drivers (sol. 1–8): IRLR3110Z DPAK (100 V, 14 mOhm, 16 mOhm at 4.5 V) driven directly by latch U5 (74HCT574, output HIGH = ON, outputs tri–stated by BLANKING → gate pull–downs hold every driver OFF). SMC 3 A tie–back diodes return to +50V. Output connector J130 (key pin 3). Supply +50V via F105 on J107–3.

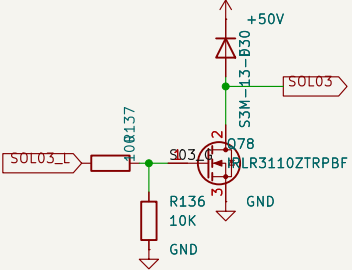
Sol 1 (high power)



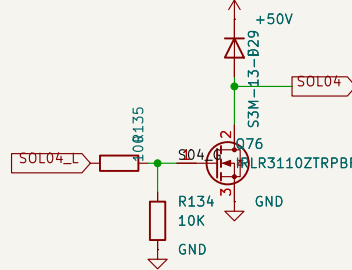
Sol 2 (high power)



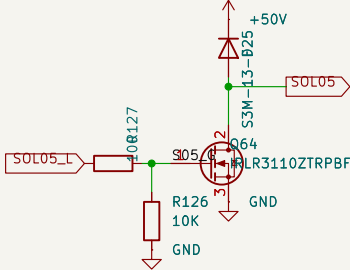
Sol 3 (high power)



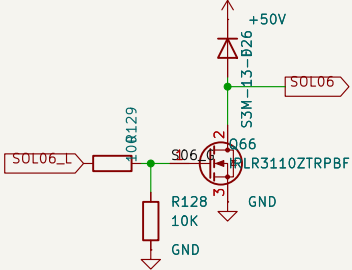
Sol 4 (high power)



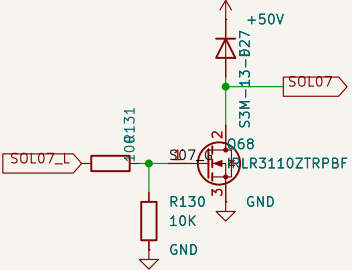
Sol 5 (high power)



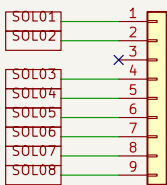
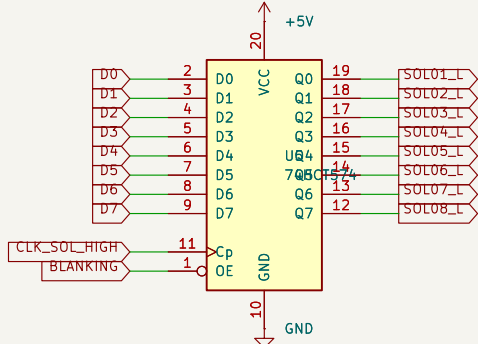
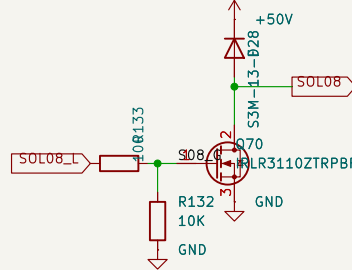
Sol 6 (high power)



Sol 7 (high power)



Sol 8 (high power)



J130 Sol 1–8 drive (playfield)

Modern re–implementation of the Williams A–12697–3 (same form factor)

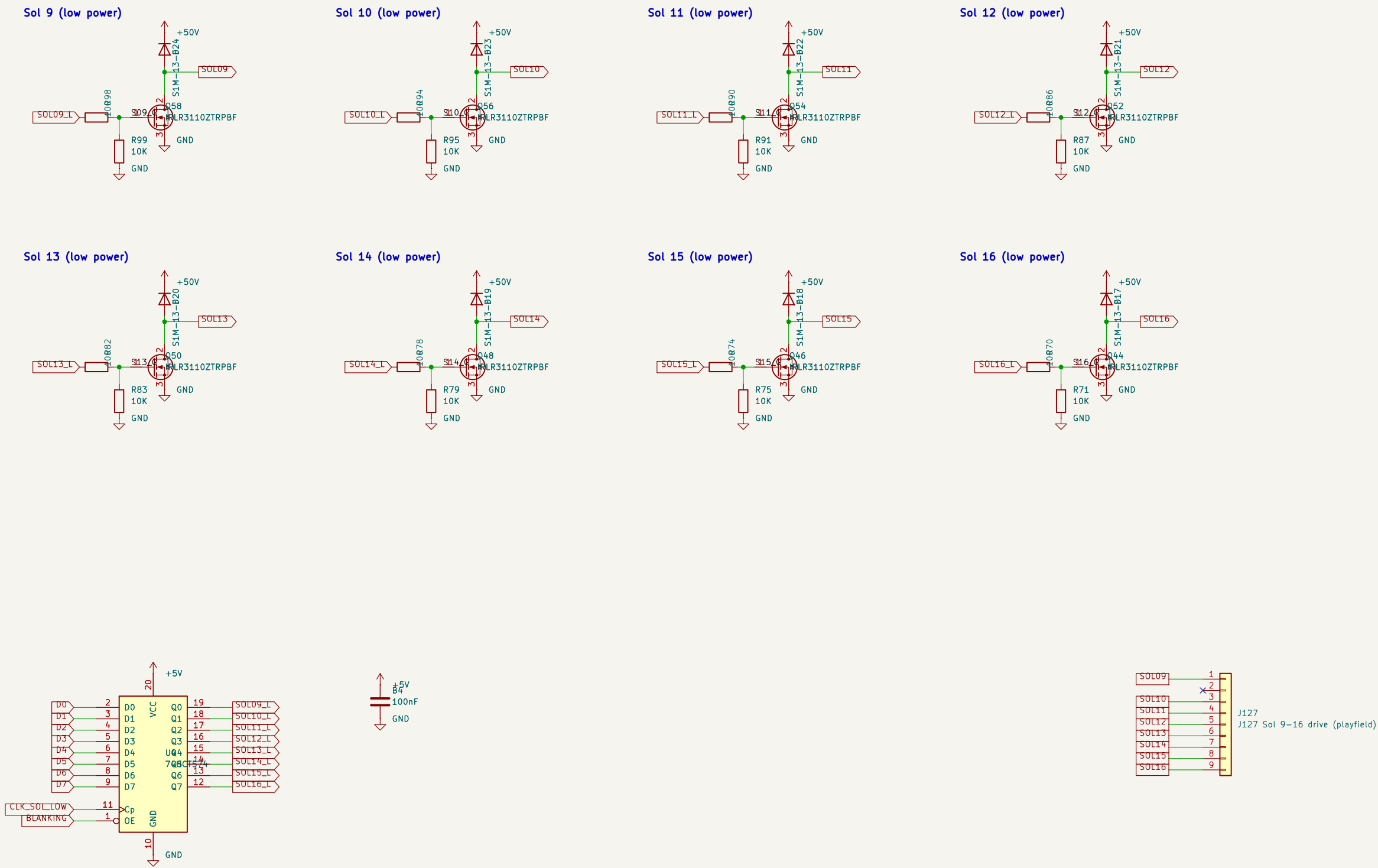
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File: sol_highpower.kicad_sch

Title: Solenoids 1–8 (high power, IRLR3110Z)

Size: A2 Date: 2026–09–04
KiCad E.D.A. 10.0.6

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Id: 4/10

Low power solenoid drivers (sol. 9–16): IRLR3110Z logic-level DPAK MOSFETs (14 mOhm), latch U4 output HIGH = ON. Tie-back diodes (SMA) to +50V. Output connector J127 (key pin 2). Supply +50V via F104 on J107–2.



Modern re-implementation of the Williams A-12697-3 (same form factor)

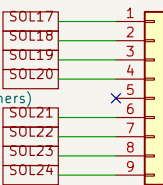
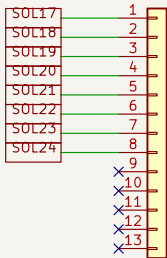
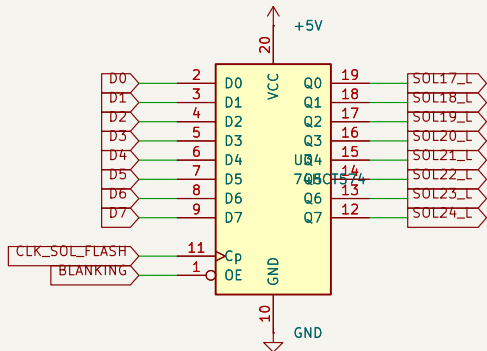
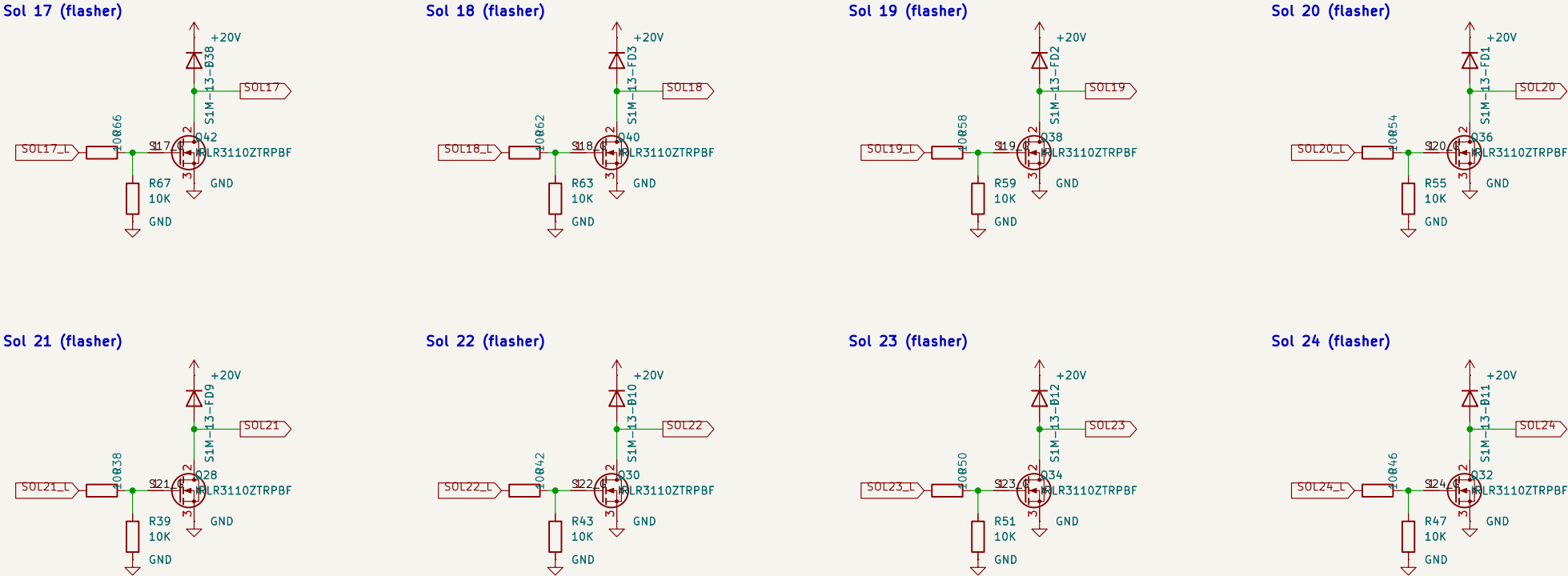
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File: sol_lowpower.kicad_sch

Title: Solenoids 9–16 (low power, IRLR3110Z)

Size: A2 Date: 2026-09-04
KiCad E.D.A. 10.0.6

Rev: A
Id: 5/10

Flashlamp drivers (sol. 17–24): IRLR3110Z DPAK, latch U3 output HIGH = lamp ON. Tie-back diodes return to +20V (flasher supply). Outputs on J126 (playfield, key pin 9) and J125 (backbox, key pin 5). Flasher supply +20V on J107–6 / J106–5.



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /Sol_Flashers/
File: sol_flashers.kicad_sch

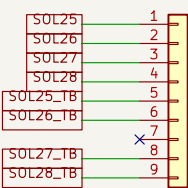
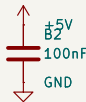
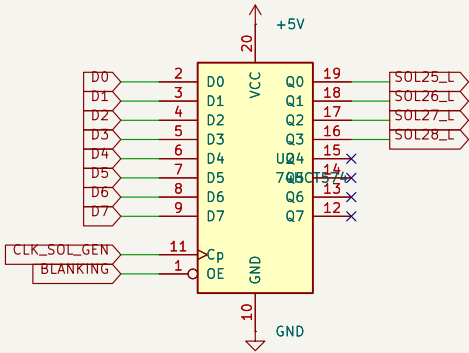
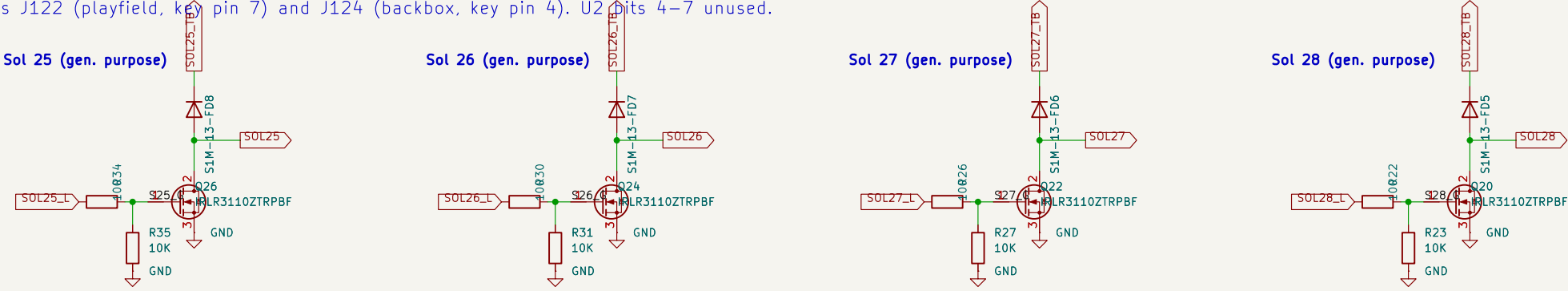
Title: Solenoids 17-24 (flashlamps, IRLR3110Z)

Size: A2
Date: 2026-09-04

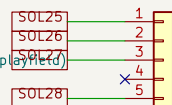
Rev: A

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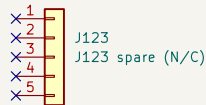
General purpose drivers (sol. 25–28): IRLR3110Z DPAK, latch U2 output HIGH = ON. Tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7) and J124 (backbox, key pin 4). U2 pins 4–7 unused.



J122 J122 Sol 25–28 drive + tie-back (playfield)



J124 J124 Sol 25–28 drive (backbox)



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /Sol_GeneralPurpose/
File: sol_gp.kicad_sch

Title: Solenoids 25–28 (general purpose, IRLR3110Z)

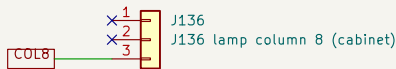
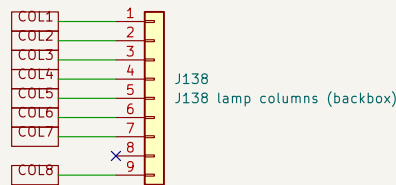
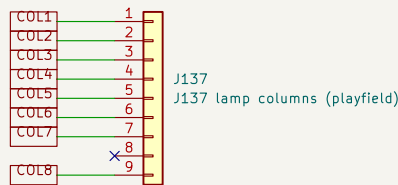
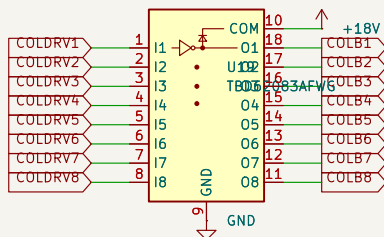
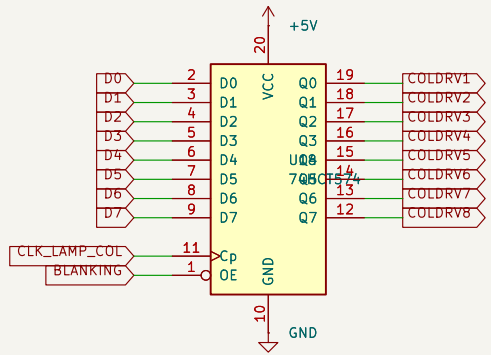
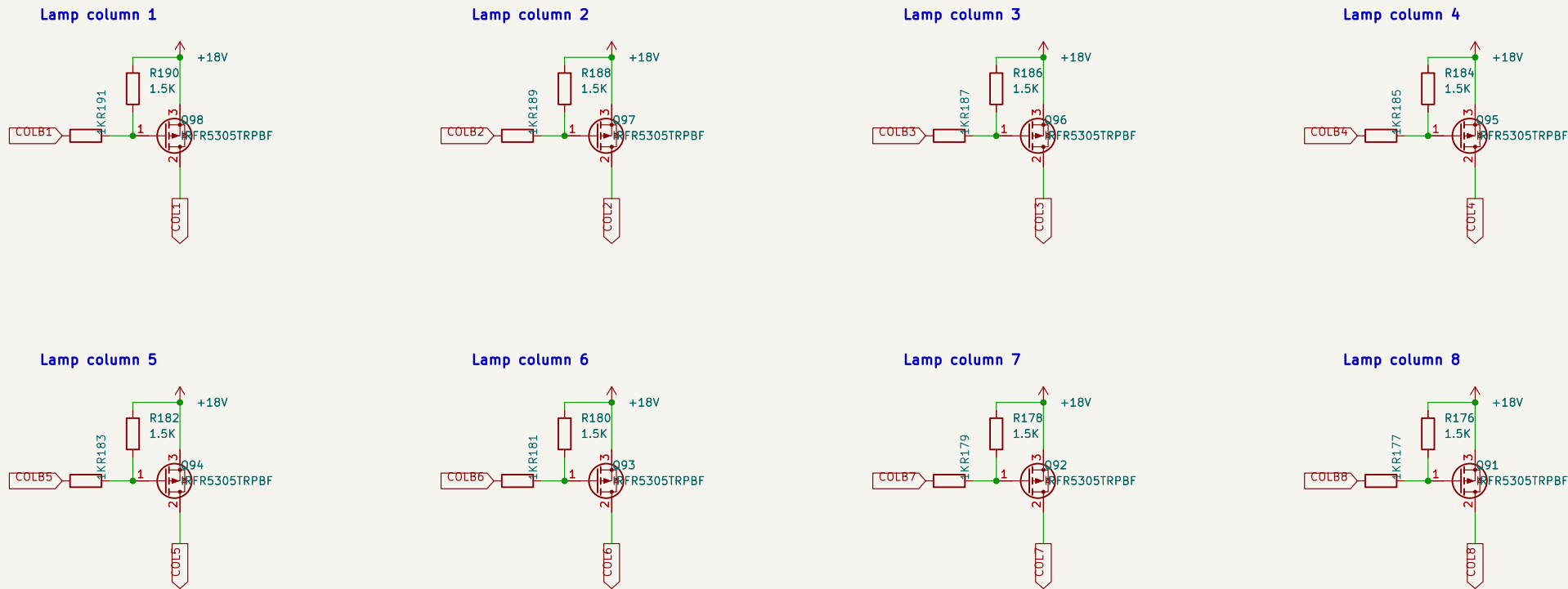
Size: A2 Date: 2026-09-04

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Rev: A

Id: 7/10

Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k.
U18 latches the data bus; the TBD62063A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor
(Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V).
Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).



Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /Lamp_Columns/
File: lamp_columns.kicad_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2 Date: 2026-09-04

KiCad E.D.A. 10.0.6

Rev: A

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General illumination: each 6.3VAC string is fused (F106–F110) on the hot side and switched by a triac in the RETURN path (MT1 to ground). Latch U1 output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (~90 mA, positive gate current in quadrants I and IV, which is why a 4–quadrant "C" grade triac is used – snubberless BW/CW types cannot fire in Q IV). 10 k base pull–downs keep the strings OFF while the latch is tri–stated by BLANKING. Bit 7 drives the optional flipper relay (DNP).

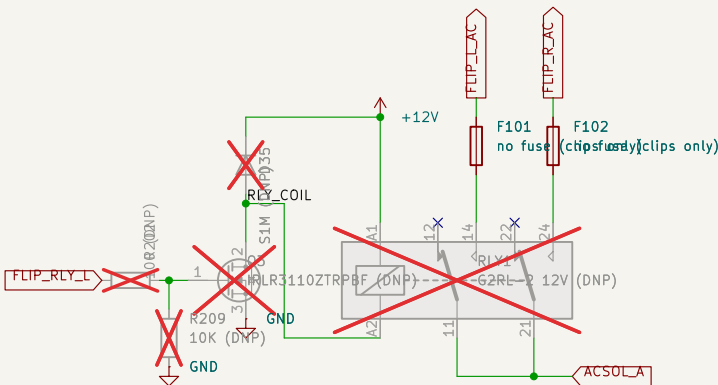
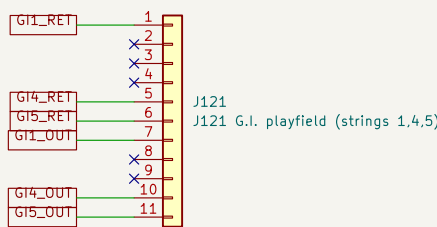
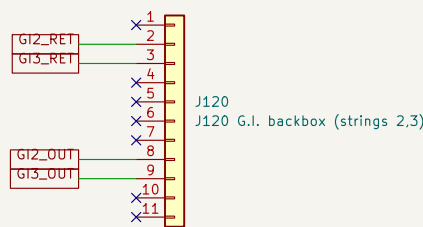
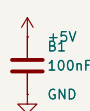
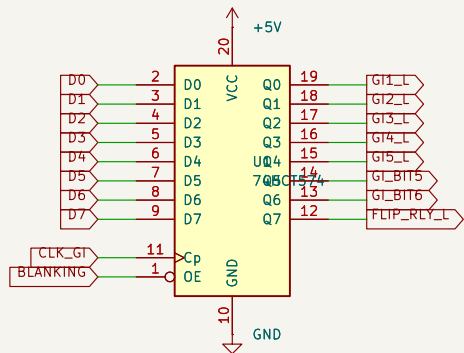
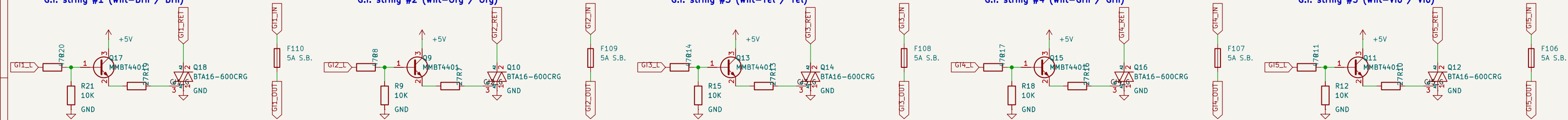
G.I. string #1 (Wht–Brn / Brn)

G.I. string #2 (Wht–Org / Org)

G.I. string #3 (Wht–Yel / Yel)

G.I. string #4 (Wht–Grn / Grn)

G.I. string #5 (Wht–Vio / Vio)



Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /GL_Triacs/
File: gl_triads.kicad_sch

Title: General Illumination triacs + flipper relay option

Size: A2 Date: 2026–09–04

KiCad E.D.A. 10.0.6

Rev: A

Id: 10/10